Maulana Azad National Institute of Technology, Bhopal
Department of Electrical Engineering
END-TERM EXAMINATION, APRIL-MAY 2024
B.Tech. IV Semester

2211301214

Subject: Electrical Machines-II
Max. Marks: 50

Subject Code: EE-222
Time: 3 Hours

Note: *All Questions are compulsory. Attempt the questions in sequence.*

Q.No.	Question	Marks	CO
1. a.	The stator of a slip-ring induction motor with slip-ring terminals open-circuited is excited from 3-phase source. The rotor is run by a prime mover. What will be the frequency of rotor induced emf at the following speeds? (a) Half synchronous speed in the same direction as the rotating magnetic field (RMF) (b) Half synchronous speed in direction opposite to RMF (c) At synchronous speed in opposite direction to RMF	03	CO1
b.	Explain with the help of suitable diagrams how rotating magnetic field is produced in a three-phase induction motor.	02	CO1
c.	A 1000 V, 50 Hz delta connected induction motor has a star connected slip ring rotor with a phase transformation ratio of 4. The rotor resistance and standstill reactance are 0.012 ohms and 0.25 ohms per phase respectively. Neglecting stator impedance and magnetizing current, determine: (i) Rotor current at start with slip rings shorted (ii) Rotor power factor at start with slip rings shorted (iii) Rotor current at 4% slip with slip rings shorted (iv) Rotor power factor at 4% slip with slip rings shorted (v) External rotor resistance per phase required to obtain a starting current of 100 A in the supply lines.	05	CO1
2. a.	Draw and explain the speed-torque characteristics of polyphase induction motor. Explain the impact of (a) Rotor resistance (b) Supply voltage on following parameters: (i) Starting torque (ii) Maximum torque (iii) Slip at maximum torque.	05	CO1
2.b.	A 1000 V, 3-phase, 50 Hz, 4-pole, star connected induction motor takes a line current of 25 A with 0.86 pf lagging. Total stator losses are 5% of the input. Rotor copper losses are 4% of the input to the rotor and mechanical losses are 3% of the input of the rotor. Calculate (i) Slip and rotor speed (ii) Torque developed in the rotor (iii) Shaft torque.	05	CO2

3. a.	i) Prove that a single-phase induction motor when excited by a single-phase supply produces two equal and opposite revolving fields. ii) Draw the speed-torque characteristic and explain why this motor does not have any starting torque?	03 02	CO4
3.b.	A 3-phase, 50 Hz, 4 pole, slip ring induction motor has a rotor resistance of 0.35 ohms/phase. The full load speed of motor is 1448 rpm. Calculate the value of external resistance per phase to be inserted in the rotor circuit to lower the speed of the motor to 1350 rpm, with the same torque.	05	CO2
4.a.	In a 3 phase, 1500 kVA, 33 kV, star connected alternator, dc resistance measured across any two terminal is 0.4 Ω. Assume effective resistance is 1.4 times the dc resistance. A field current of 10 A generated 263 A armature current during short circuit test and the same field current resulted 1100 V during open circuit test. Calculate synchronous reactance and the voltage regulation at full load 0.8 pf lagging.	06	CO5
4.b.	A 4 pole, 3 phase, 50 Hz, star connected alternator has 60 slots with 4 conductors per slot. Coils are short pitched by 3 slots. If all the turns per phase are connected in series and the phase spread is 60 degree, find the line voltage induced for a flux per pole of 0.943 Wb distributed sinusoidally in space.	04	CO3
5.a.	A 1600 KVA, 13.5 kV, three phase, star connected alternator has the per phase effective armature resistance and synchronous reactance of 1.5 and 30 ohms, respectively. Calculate the percentage regulation for a load of 128 kW operating at power factor of (i) 0.8 Leading, and (ii) 0.6 lagging.	06	CO3
5.b.	Two generators rated at 200 MW and 400 MW are operating in parallel. Their governor droop characteristics are respectively 4% and 5% from no load to full load. At no load the system frequency is 50 Hertz when supplying a load of 600 MW. Find the system frequency.	04	CO3

$f_{L} - f \Rightarrow (1 - droop)P$

$\Rightarrow (1 - 0.04)\, 50$

$\Rightarrow 47.5$

$\frac{200}{600}$ f

47.5 48 48